

For infinitesimal changes in r , Diamond and Stiglitz use the differential versions of properties (i) and (iii) to rank distributions with the same mean in order of riskiness. An increase in r is said to represent a mean-preserving increase in risk if

$$(iv) \quad \int_0^B F_r(\theta, r) d\theta = 0$$

$$(v) \quad \int_0^y F_r(\theta, r) d\theta \geq 0, \quad 0 \leq y \leq B,$$

where $F_r(\theta, r)$

An example of Stochastic dynamic programming. To be discussed in class

6.3. McCall's model of intertemporal job search

We now consider an unemployed worker who is searching for a job under the following circumstances: Each period the worker draws one offer w from the same wage distribution $F(W) = \text{prob}\{w \leq W\}$, with $F(0) = 0$, $F(B) = 1$ for $B < \infty$. The worker has the option of rejecting the offer, in which case he or she receives c this period in unemployment compensation and waits until next period to draw another offer from F ; alternatively, the worker can accept the offer to work at w , in which case he or she receives a wage of w per period forever. Neither quitting nor firing is permitted.

Let y_t be the worker's income in period t . We have $y_t = c$ if the worker is unemployed and $y_t = w$ if the worker has accepted an offer to work at wage w . The unemployed worker devises a strategy to maximize $E \sum_{t=0}^{\infty} \beta^t y_t$ where $0 < \beta < 1$ is a discount factor.

Let $v(w)$ be the expected value of $\sum_{t=0}^{\infty} \beta^t y_t$ for a worker who has offer w in hand, who is deciding whether to accept or to reject it, and who behaves optimally. We assume no recall. The value function $v(w)$ satisfies the Bellman equation

$$v(w) = \max \left\{ \frac{w}{1-\beta}, c + \beta \int v(w') dF(w') \right\}, \quad (6.3.1)$$

distribution $F(\theta, r_1)$. A definition based on (i) and (iii), however, does provide a transitive ordering, which is a desirable feature for a definition designed to order distributions according to their riskiness.

where the maximization is over the two actions: (1) *accept* the wage offer w and work forever at wage w , or (2) *reject* the offer, receive c this period, and draw a new offer w' from distribution F next period. Fig. 6.3.1 graphs the functional equation (6.3.1) and reveals that its solution will be of the form

$$v(w) = \begin{cases} \frac{\bar{w}}{1-\beta} = c + \beta \int_0^B v(w') dF(w') & \text{if } w \leq \bar{w} \\ \frac{w}{1-\beta} & \text{if } w \geq \bar{w}. \end{cases} \quad (6.3.2)$$

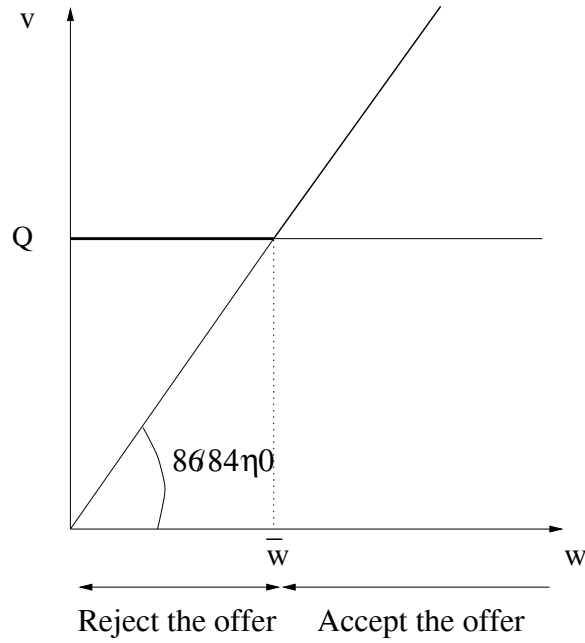


Figure 6.3.1: The function $v(w) = \max\{w/(1-\beta), c + \beta \int_0^B v(w') dF(w')\}$. The reservation wage $\bar{w} = (1-\beta)[c + \beta \int_0^B v(w') dF(w')]$.

Using equation (6.3.2), we can convert the functional equation (6.3.1) into an ordinary equation in the reservation wage $\bar{w}$. Evaluating $v(\bar{w})$ and using

equation (6.3.2), we have

$$\frac{\bar{w}}{1-\beta} = c + \beta \int_0^{\bar{w}} \frac{\bar{w}}{1-\beta} dF(w') + \beta \int_{\bar{w}}^B \frac{w'}{1-\beta} dF(w')$$

or

$$\begin{aligned} \frac{\bar{w}}{1-\beta} \int_0^{\bar{w}} dF(w') + \frac{\bar{w}}{1-\beta} \int_{\bar{w}}^B dF(w') \\ = c + \beta \int_0^{\bar{w}} \frac{\bar{w}}{1-\beta} dF(w') + \beta \int_{\bar{w}}^B \frac{w'}{1-\beta} dF(w') \end{aligned}$$

or

$$\bar{w} \int_0^{\bar{w}} dF(w') - c = \frac{1}{1-\beta} \int_{\bar{w}}^B (\beta w' - \bar{w}) dF(w').$$

Adding $\bar{w} \int_{\bar{w}}^B dF(w')$ to both sides gives

$$(\bar{w} - c) = \frac{\beta}{1-\beta} \int_{\bar{w}}^B (w' - \bar{w}) dF(w'). \quad (6.3.3)$$

Equation (6.3.3) is often used to characterize the determination of the reservation wage $\bar{w}$. The left side is the cost of searching one more time when an offer $\bar{w}$ is in hand. The right side is the expected benefit of searching one more time in terms of the expected present value associated with drawing $w' > \bar{w}$. Equation (6.3.3) instructs the agent to set $\bar{w}$ so that the cost of searching one more time equals the benefit.

Let us define the function on the right side of equation (6.3.3) as

$$h(w) = \frac{\beta}{1-\beta} \int_w^B (w' - w) dF(w'). \quad (6.3.4)$$

Notice that $h(0) = Ew\beta/(1-\beta)$, that $h(B) = 0$, and that $h(w)$ is differentiable, with derivative given by³

$$h'(w) = -\frac{\beta}{1-\beta} [1 - F(w)] < 0.$$

³ To compute $h'(w)$, we apply Leibniz' rule to equation (6.3.4). Let $\phi(t) = \int_{\alpha(t)}^{\beta(t)} f(x, t) dx$ for $t \in [c, d]$. Assume that f and f_t are continuous and that α, β are differentiable on $[c, d]$. Then Leibniz' rule asserts that $\phi(t)$ is differentiable on $[c, d]$ and

$$\phi'(t) = f[\beta(t), t] \beta'(t) - f[\alpha(t), t] \alpha'(t) + \int_{\alpha(t)}^{\beta(t)} f_t(x, t) dx.$$

To apply this formula to the equation in the text, let w play the role of t .

We also have

$$h''(w) = \frac{\beta}{1-\beta} F'(w) > 0,$$

so that $h(w)$ is convex to the origin. Fig. 6.3.2 graphs $h(w)$ against $(w-c)$ and indicates how $\bar{w}$ is determined. From Figure 5.3 it is apparent that an increase in c leads to an increase in $\bar{w}$.

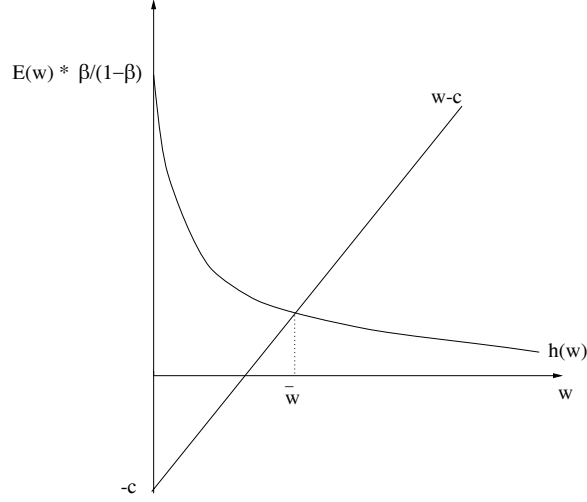


Figure 6.3.2: The reservation wage, $\bar{w}$, that satisfies $\bar{w} - c = [\beta/(1-\beta)] \int_{\bar{w}}^B (w' - \bar{w}) dF(w') \equiv h(\bar{w})$.

To get an alternative characterization of the condition determining $\bar{w}$, we return to equation (6.3.3) and express it as

$$\begin{aligned} \bar{w} - c &= \frac{\beta}{1-\beta} \int_{\bar{w}}^B (w' - \bar{w}) dF(w') + \frac{\beta}{1-\beta} \int_0^{\bar{w}} (w' - \bar{w}) dF(w') \\ &\quad - \frac{\beta}{1-\beta} \int_0^{\bar{w}} (w' - \bar{w}) dF(w') \\ &= \frac{\beta}{1-\beta} Ew - \frac{\beta}{1-\beta} \bar{w} - \frac{\beta}{1-\beta} \int_0^{\bar{w}} (w' - \bar{w}) dF(w') \end{aligned}$$

or

$$\bar{w} - (1-\beta)c = \beta Ew - \beta \int_0^{\bar{w}} (w' - \bar{w}) dF(w').$$

Applying integration by parts to the last integral on the right side and rearranging, we have

$$\bar{w} - c = \beta(Ew - c) + \beta \int_0^{\bar{w}} F(w') dw'. \quad (6.3.5)$$

At this point it is useful to define the function

$$g(s) = \int_0^s F(p) dp. \quad (6.3.6)$$

This function has the characteristics that $g(0) = 0$, $g(s) \geq 0$, $g'(s) = F(s) > 0$, and $g''(s) = F'(s) > 0$ for $s > 0$. Then equation (6.3.5) can be expressed alternatively as $\bar{w} - c = \beta(Ew - c) + \beta g(\bar{w})$, where $g(s)$ is the function defined by equation (6.3.6). In Figure 5.4 we graph the determination of $\bar{w}$, using equation (6.3.5).

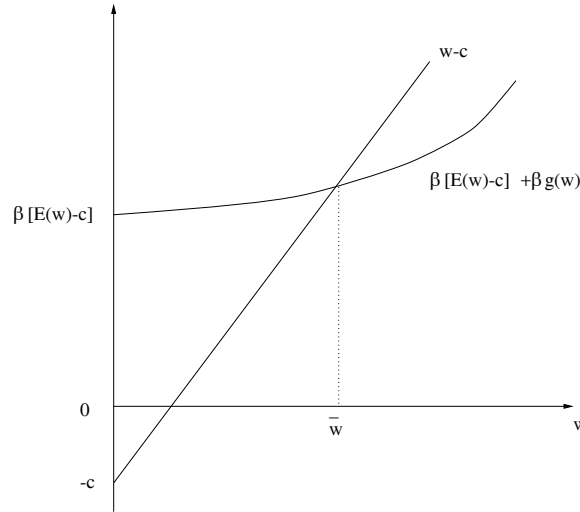


Figure 6.3.3: The reservation wage, $\bar{w}$, that satisfies $\bar{w} - c = \beta(Ew - c) + \beta \int_0^{\bar{w}} F(w') dw' \equiv \beta(Ew - c) + \beta g(\bar{w})$.

That is, the mean waiting time to a successful job offer equals the reciprocal of the probability of an accepted offer.

We invite the reader to prove that the mean waiting time increases with increases in the rate of unemployment compensation, c .

This subsection is
discussed in class

6.3.4. Firing

We now briefly consider a modification of the job search model in which each period after the first period on the job the worker faces probability α of being fired, where $1 > \alpha > 0$. The probability α of being fired next period is assumed to be independent of tenure. The worker continues to sample wage offers from a time-invariant and known probability distribution F and to receive unemployment compensation in the amount c . The worker receives a time-invariant wage w on a job until she is fired. A worker who is fired becomes unemployed for one period before drawing a new wage.

We let $v(w)$ be the expected present value of income of a previously unemployed worker who has offer w in hand and who behaves optimally. If she rejects the offer, she receives c in unemployment compensation this period and next period draws a new offer w' , whose value to her now is $\beta \int v(w') dF(w')$. If she rejects the offer, $v(w) = c + \beta \int v(w') dF(w')$. If she accepts the offer, she receives w this period, with probability $1 - \alpha$ that she is not fired next period, in which case she receives $\beta v(w)$ and with probability α that she is fired, and after one period of unemployment draws a new wage, receiving $\beta[c + \beta \int v(w') dF(w')]$. Therefore, if she accepts the offer, $v(w) = w + \beta(1 - \alpha)v(w) + \beta\alpha[c + \beta \int v(w') dF(w')]$. Thus the Bellman equation becomes

$$v(w) = \max\{w + \beta(1 - \alpha)v(w) + \beta\alpha[c + \beta \int v(w') dF(w')], c + \beta \int v(w') dF(w')\},$$

⁴ An alternative way of deriving the mean waiting time is to use the algebra of z transforms, we say that $h(z) = \sum_{j=0}^{\infty} h_j z^j$ and note that $h'(z) = \sum_{j=1}^{\infty} j h_j z^{j-1}$ and $h'(1) = \sum_{j=1}^{\infty} j h_j$. (For an introduction to z transforms, see Gabel and Roberts, 1973.) The z transform of the sequence $(1 - \lambda)\lambda^{j-1}$ is given by $\sum_{j=1}^{\infty} (1 - \lambda)\lambda^{j-1} z^j = (1 - \lambda)z/(1 - \lambda z)$. Evaluating $h'(z)$ at $z = 1$ gives, after some simplification, $h'(1) = 1/(1 - \lambda)$. Therefore we have that the mean waiting time is given by $(1 - \lambda) \sum_{j=1}^{\infty} j \lambda^{j-1} = 1/(1 - \lambda)$.

where $Ev = \int v(w')dF(w')$. This equation has a solution of the form⁵

$$v(w) = \begin{cases} \frac{w + \beta\alpha[c + \beta Ev]}{1 - \beta(1 - \alpha)}, & \text{if } w \geq \bar{w} \\ c + \beta Ev, & w \leq \bar{w} \end{cases}$$

where $\bar{w}$ solves

$$\frac{\bar{w} + \beta\alpha[c + \beta Ev]}{1 - \beta(1 - \alpha)} = c + \beta Ev. \quad (6.3.7)$$

The optimal policy is of the reservation wage form. The reservation wage $\bar{w}$ will not be characterized here as a function of c , F , and α ; the reader is invited to do so by pursuing the implications of the preceding formula.

6.4. A lake model

Consider an economy consisting of a continuum of *ex ante* identical workers living in the environment described in the previous section. These workers move recurrently between unemployment and employment. The mean duration of each spell of employment

is $\frac{1}{\alpha}$ and the mean duration of unemployment is $\frac{1}{1 - F(\bar{w})}$. The average unemployment rate U_t across the continuum of workers obeys the difference equation

$$U_{t+1} = \alpha(1 - U_t) + F(\bar{w})U_t,$$

where α is the hazard rate of escaping employment and $[1 - F(\bar{w})]$ is the hazard rate of escaping unemployment. Solving this difference equation for a stationary solution, i.e., imposing $U_{t+1} = U_t = U$, gives $U = \frac{\alpha}{\alpha + 1 - F(\bar{w})}$ or

$$U = \frac{\frac{1}{1 - F(\bar{w})}}{\frac{1}{\alpha} + \frac{1}{1 - F(\bar{w})}}. \quad (6.4.1)$$

Equation (6.4.1) expresses the stationary unemployment rate in terms of the ratio of the average duration of unemployment to the sum of average durations

⁵ That it takes this form can be established by guessing that $v(w)$ is nondecreasing in w . This guess implies the equation in the text for $v(w)$, which is nondecreasing in w . This argument verifies that $v(w)$ is nondecreasing, given the uniqueness of the solution of the Bellman equation.